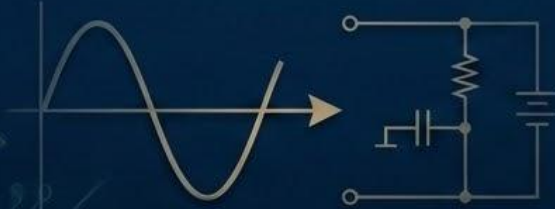


# DEVRE ANALİZİ II

**Hafta 2** SINÜZOİDAL SÜREKLİ DURUM ANALİZİ



Find the phasors corresponding to the following signals:

(a)  $v(t) = 21 \cos(4t - 15^\circ) \text{ V}$

(b)  $i(t) = -8 \sin(10t + 70^\circ) \text{ mA}$

(c)  $v(t) = 120 \sin(10t - 50^\circ) \text{ V}$

(d)  $i(t) = -60 \cos(30t + 10^\circ) \text{ mA}$

Transform the following sinusoids to phasors:

(a)  $-10 \cos(4t + 75^\circ)$       (b)  $5 \sin(20t - 10^\circ)$

(c)  $4 \cos 2t + 3 \sin 2t$

Obtain the sinusoids corresponding to each of the following phasors:

(a)  $\mathbf{V}_1 = 60 \angle 15^\circ \text{ V}, \omega = 1$

(b)  $\mathbf{V}_2 = 6 + j8 \text{ V}, \omega = 40$

(c)  $\mathbf{I}_1 = 2.8e^{-j\pi/3} \text{ A}, \omega = 377$

(d)  $\mathbf{I}_2 = -0.5 - j1.2 \text{ A}, \omega = 10^3$

Apply phasor analysis to evaluate the following.

(a)  $v = 50 \cos(\omega t + 30^\circ) + 30 \cos(\omega t - 90^\circ) \text{ V}$

(b)  $i = 15 \cos(\omega t + 45^\circ) - 10 \sin(\omega t + 45^\circ) \text{ A}$

Calculate these complex numbers and express your results in rectangular form:

$$(a) \frac{15 \angle 45^\circ}{3 - j4} + j2$$

$$(b) \frac{8 \angle -20^\circ}{(2 + j)(3 - j4)} + \frac{10}{-5 + j12}$$

$$(c) 10 + (8 \angle 50^\circ)(5 - j12)$$

Evaluate the following complex numbers and leave your results in polar form:

(a)  $5\angle 30^\circ \left( 6 - j8 + \frac{3\angle 60^\circ}{2 + j} \right)$

(b)  $\frac{(10\angle 60^\circ)(35\angle -50^\circ)}{(2 + j6) - (5 + j)}$

Evaluate the following complex numbers:

$$(a) \frac{2 + j3}{1 - j6} + \frac{7 - j8}{-5 + j11}$$

$$(b) \frac{(5 \angle 10^\circ)(10 \angle -40^\circ)}{(4 \angle -80^\circ)(-6 \angle 50^\circ)}$$

$$(c) \begin{vmatrix} 2 + j3 & -j2 \\ -j2 & 8 - j5 \end{vmatrix}$$

Evaluate these determinants:

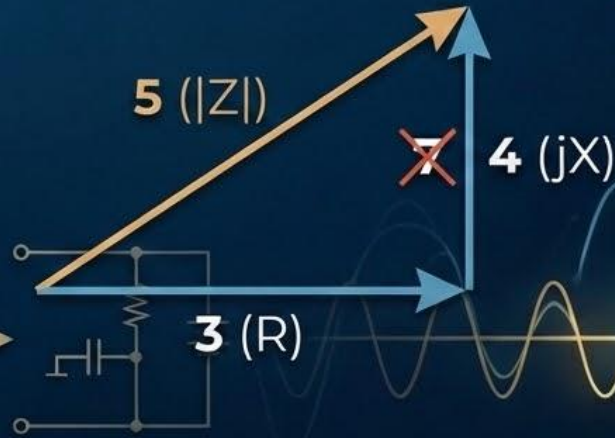
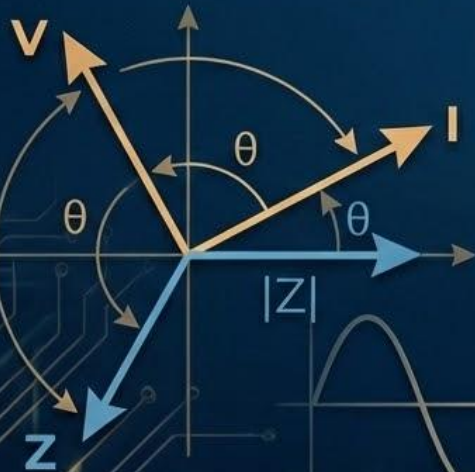
$$(a) \begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix}$$

$$(b) \begin{vmatrix} 20 \angle -30^\circ & -4 \angle -10^\circ \\ 16 \angle 0^\circ & 3 \angle 45^\circ \end{vmatrix}$$

$$(c) \begin{vmatrix} 1 - j & -j & 0 \\ j & 1 & -j \\ 1 & j & 1 + j \end{vmatrix}$$

$$V = I \cdot Z$$

3+4 artık 7 etmeyebilir

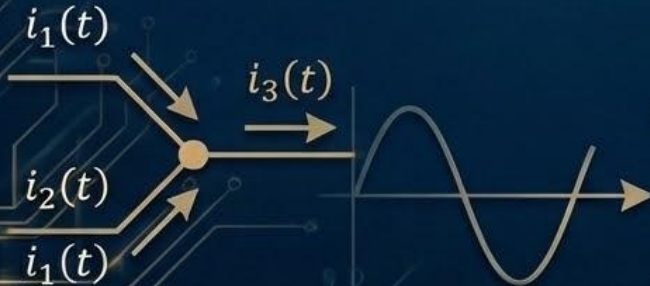


# KCL (Akımlar Yasası)

$$\sum I_n = 0$$

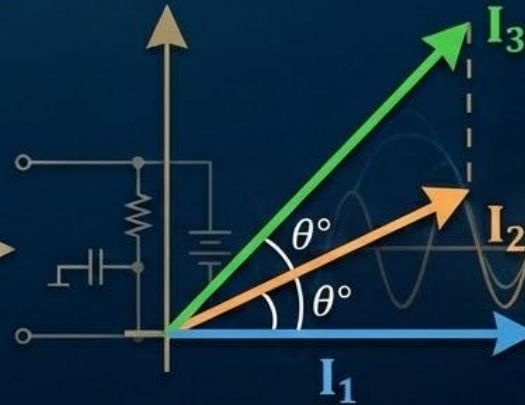
## Zaman Tanım Kümesi:

**Örnek:** Bir düğüme giren  $i_1(t)$  ve  $i_2(t)$  olsun. Çıkan  $i_3(t)$ .



## Fazör Tanım Kümesi:

Fazör:  $I_1 + I_2 = I_3$

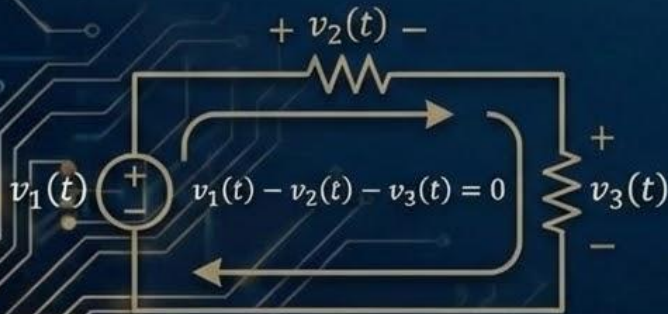


# KVL (Gerilimler Yasası)

$$\sum V_n = 0$$

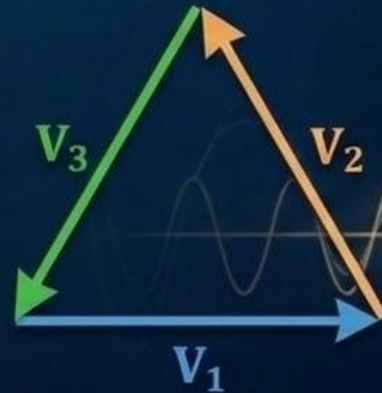
## Zaman Tanım Kümesi:

**Örnek:** Kapalı bir döngüdeki gerilimlerin toplamı sıfırdır.



## Fazör Tanım Kümesi:

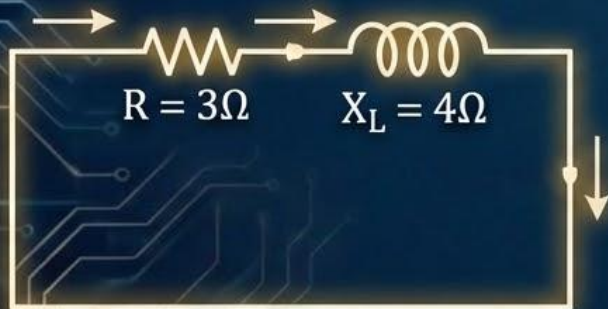
Fazör:  $V_1 + V_2 + V_3 = 0$



# Seri Bağlantı (Series Connection)

$$Z_{es} = Z_1 + Z_2 + \dots$$

Şekil (Figure):



Formül (Formula):

Hesap (Calculation):

$$Z_1 = 3\angle 0^\circ$$

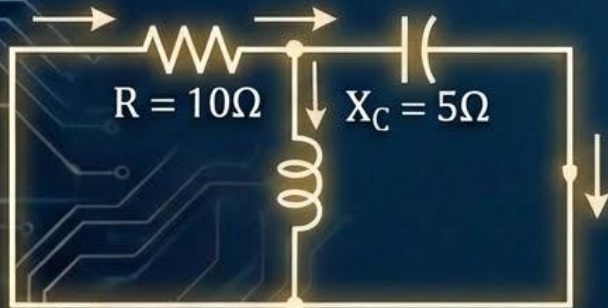
$$Z_2 = j4 = 4\angle 90^\circ$$

$$Z_{es} = 3 + j4 = 5\angle 53.1^\circ \Omega$$

## 2. Paralel Bağlantı: (Parallel Connection)

$$\frac{1}{Z_{es}} = \frac{1}{Z_1} + \frac{1}{Z_2}, \quad Z_{es} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

**Şekil (Figure):**



**Hesap (Calculation):**

$$Z_1 = 10\Omega, \quad Z_2 = -j5\Omega$$

$$Z_{es} = \frac{10 \cdot (-j5)}{10 - j5} = \frac{-j50}{10 - j5}$$

**Eşlenik ile Çözüm (Conjugate Solution):**

Paydayı eşleniğiyle  $(10 + j5)$  çarpalım  
(Multiply denominator by conjugate)

**Sonuç (Result):**  $2 - j4\Omega$

# D. Y - Δ (Yıldız-Üçgen) Dönüşümü

- **Hatırlatma:**

"DC devrelerdeki formüllerin aynısıdır. Sadece R yerine Z yazacaksınız."

- **Δ → Y Formülü:**

$$Z_a = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

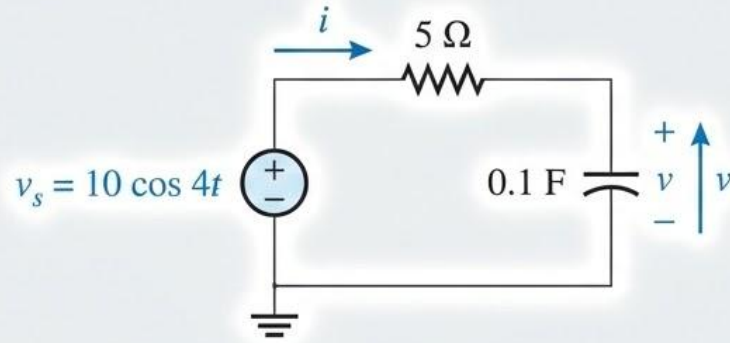
- **Uygulama Alanı:**

"Bunu özellikle 3-Fazlı devrelerde ve köprü devrelerinde çok kullanacağız,

(Yıldızın bir kolu = Ona komşu iki Üçgen kolunun çarpımı / Toplam Empedans)

- **Uygulama Alanı:** "Bunu özellikle 3-Fazlı devrelerde ve köprü devrelerinde çok kullanacağız, formülü not edin."

Şekilde gösterilen devrede  $v(t)$  ve  $i(t)$  değerlerini bulunuz.



**Solution:**

From the voltage source  $10 \cos 4t$ ,  $\omega = 4$ ,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \, \Omega$$

Hence the current

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10 \angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789 \angle 26.57^\circ \text{ A} \end{aligned} \quad (9.9.1)$$

The voltage across the capacitor is

$$\begin{aligned} \mathbf{V} = \mathbf{I} \mathbf{Z}_C &= \frac{\mathbf{I}}{j\omega C} = \frac{1.789 \angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789 \angle 26.57^\circ}{0.4 \angle 90^\circ} = 4.47 \angle -63.43^\circ \text{ V} \end{aligned} \quad (9.9.2)$$

Converting  $\mathbf{I}$  and  $\mathbf{V}$  in Eqs. (9.9.1) and (9.9.2) to the time domain, we get

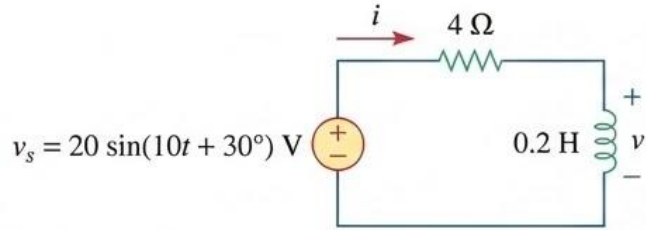
$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

Notice that  $i(t)$  leads  $v(t)$  by  $90^\circ$  as expected.

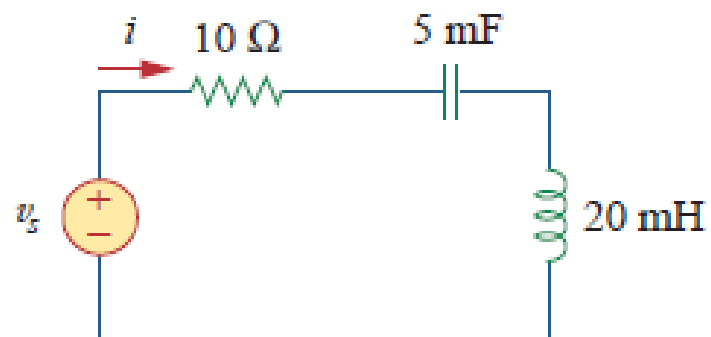
**Fig**  
For

# $v(t)$ ve $i(t)$ değerlerini bulunuz.

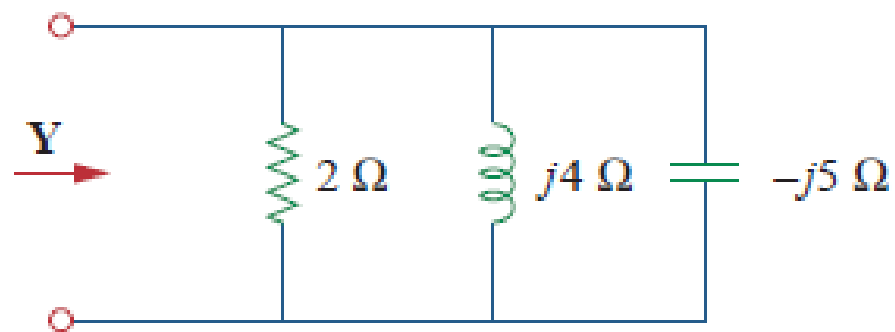


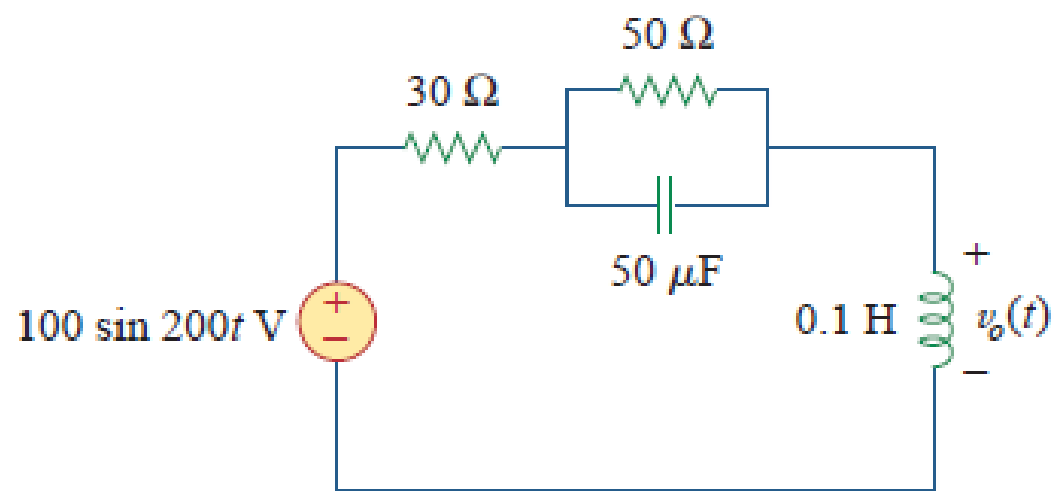
$$8.944 \sin(10t + 93.43^\circ) \text{ V}, 4.472 \sin(10t + 3.43^\circ) \text{ A}.$$

Find current  $i$  in the circuit of Fig. 9.42, when  $v_s(t) = 50 \cos 200t$  V.

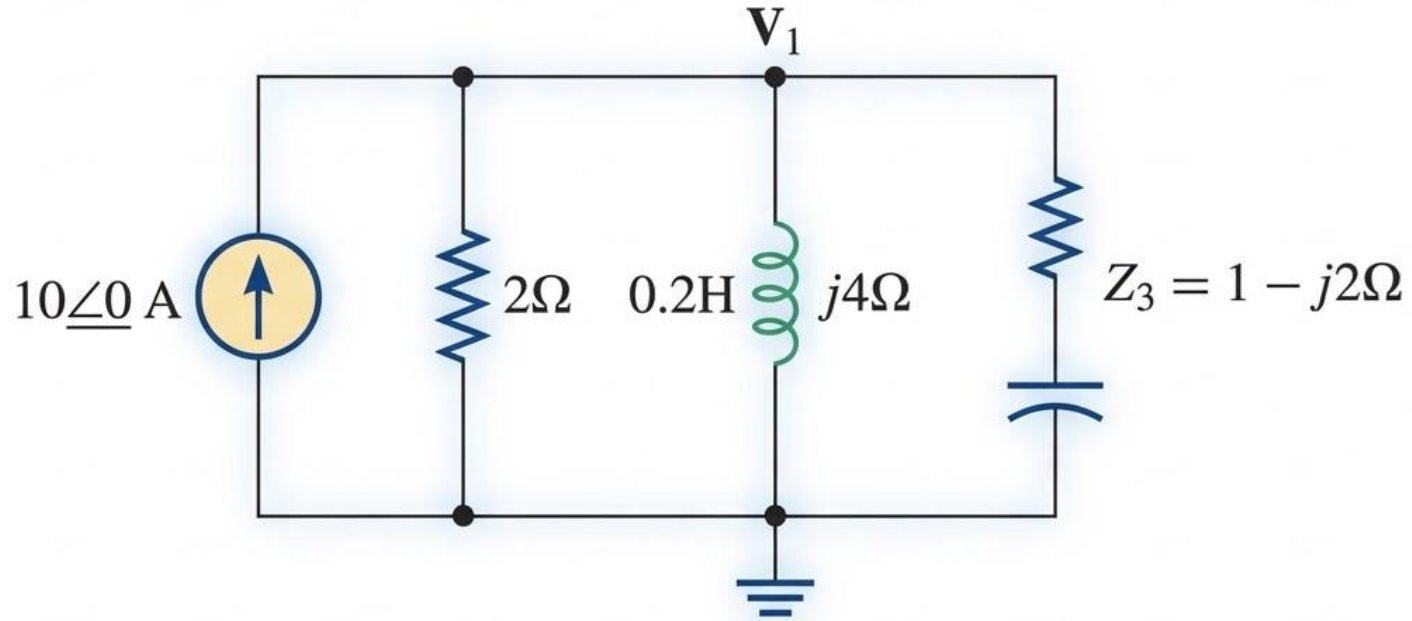


Determine the admittance  $\mathbf{Y}$  for the circuit in Fig.

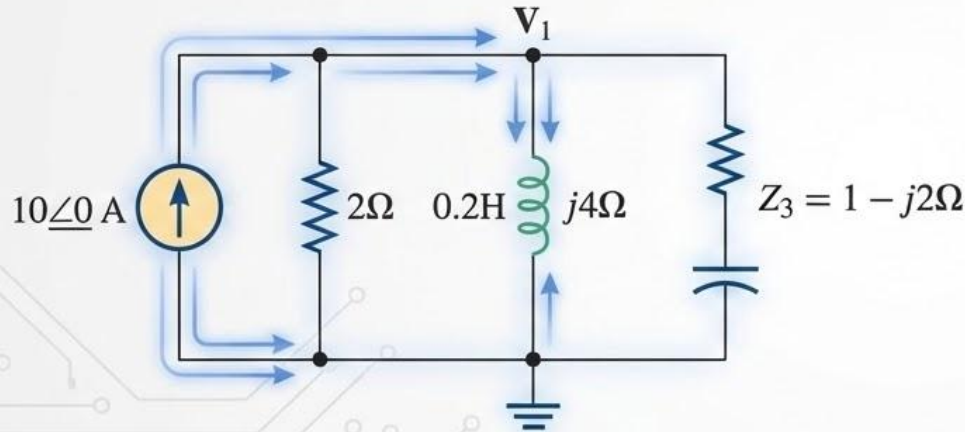




## DÜĞÜM VE ÇEVRE ANALİZİ



# Giren Akım = Çıkan Akımlar



$$10\angle 0 = \frac{V_1}{2} + \frac{V_1}{j4} + \frac{V_1}{1-j2}$$

Admisanslar ( $Y = 1/Z$ )

$$Y_1 = 0.5 \text{ S}$$

$$Y_2 = 1/j4 = -j0.25 \text{ S}$$

$$Y_3 = \frac{1}{1-j2} = \frac{1+j2}{5} = 0.2 + j0.4 \text{ S}$$

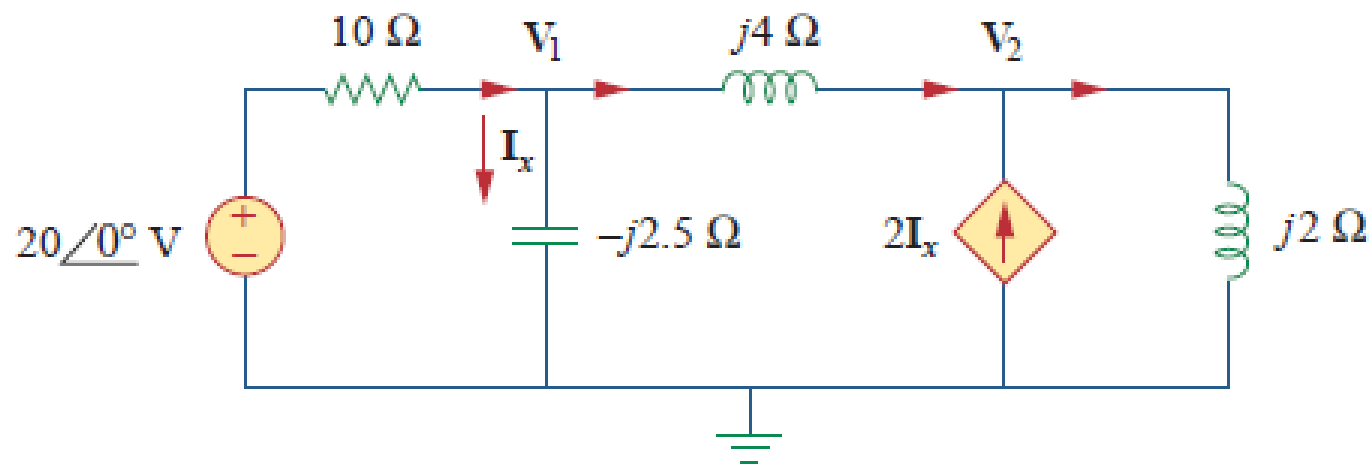
Denklemin Basitleştirilmesi

$$10 = V_1(0.5 - j0.25 + 0.2 + j0.4)$$

$$10 = V_1(0.7 + j0.15)$$

Düğüm Gerilimi  $V_1$  Çözümü

$$V_1 = \frac{10}{0.7 + j0.15} \approx 13.96\angle -12.1^\circ \text{ V}$$



Applying KCL at node 1,

$$\frac{20 - \mathbf{V}_1}{10} = \frac{\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4}$$

or

$$(1 + j1.5)\mathbf{V}_1 + j2.5\mathbf{V}_2 = 20 \quad (10.1.1)$$

At node 2,

$$2\mathbf{I}_x + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

But  $\mathbf{I}_x = \mathbf{V}_1 / -j2.5$ . Substituting this gives

$$\frac{2\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

By simplifying, we get

$$11\mathbf{V}_1 + 15\mathbf{V}_2 = 0 \quad (10.1.2)$$

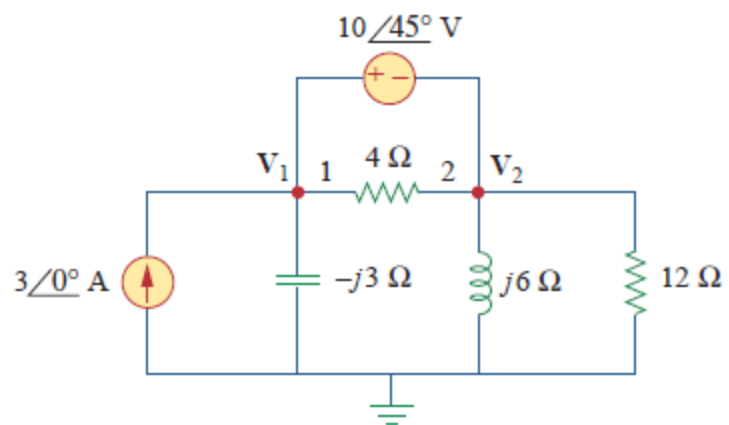
Equations (10.1.1) and (10.1.2) can be put in matrix form as

$$\begin{bmatrix} 1 + j1.5 & j2.5 \\ 11 & 15 \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

We obtain the determinants as

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 + j1.5 & j2.5 \\ 11 & 15 \end{vmatrix} = 15 - j5 \\ \Delta_1 &= \begin{vmatrix} 20 & j2.5 \\ 0 & 15 \end{vmatrix} = 300, & \Delta_2 &= \begin{vmatrix} 1 + j1.5 & 20 \\ 11 & 0 \end{vmatrix} = -220 \\ \mathbf{V}_1 &= \frac{\Delta_1}{\Delta} = \frac{300}{15 - j5} = 18.97 \angle 18.43^\circ \text{ V} \\ \mathbf{V}_2 &= \frac{\Delta_2}{\Delta} = \frac{-220}{15 - j5} = 13.91 \angle 198.3^\circ \text{ V} \end{aligned}$$

Compute  $V_1$  and  $V_2$  in the circuit of Fig. 10.4.



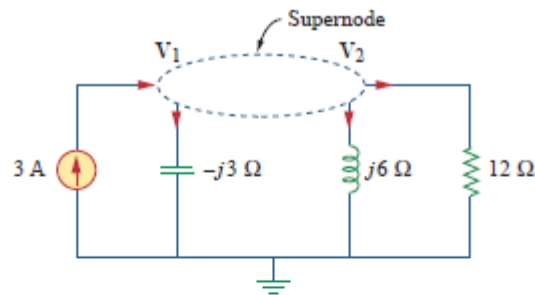
**Solution:**

Nodes 1 and 2 form a supernode as shown in Fig. 10.5. Applying KCL at the supernode gives

$$3 = \frac{V_1}{-j3} + \frac{V_2}{j6} + \frac{V_2}{12}$$

or

$$36 = j4V_1 + (1 - j2)V_2 \quad (10.2.1)$$



**Figure 10.5**

A supernode in the circuit of Fig. 10.4.

But a voltage source is connected between nodes 1 and 2, so that

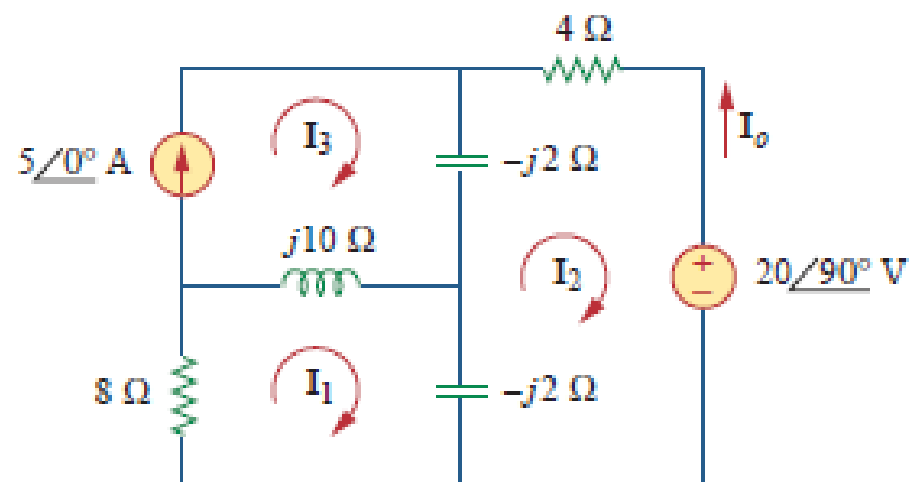
$$V_1 = V_2 + 10\angle 45^\circ \quad (10.2.2)$$

Substituting Eq. (10.2.2) in Eq. (10.2.1) results in

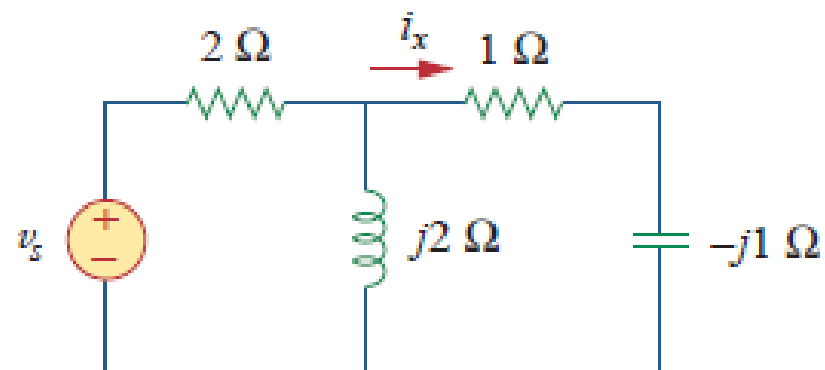
$$36 - 40\angle 135^\circ = (1 + j2)V_2 \Rightarrow V_2 = 31.41\angle -87.18^\circ \text{ V}$$

From Eq. (10.2.2),

$$V_1 = V_2 + 10\angle 45^\circ = 25.78\angle -70.48^\circ \text{ V}$$



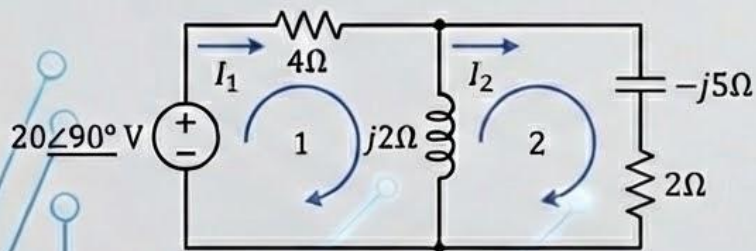
Find  $v_s(t)$  in the circuit of Fig. 9.56 if the current  $i_x$  through the  $1\text{-}\Omega$  resistor is  $0.5 \sin 200t$  A.



## B. Çevre Akımları (Mesh Analysis)

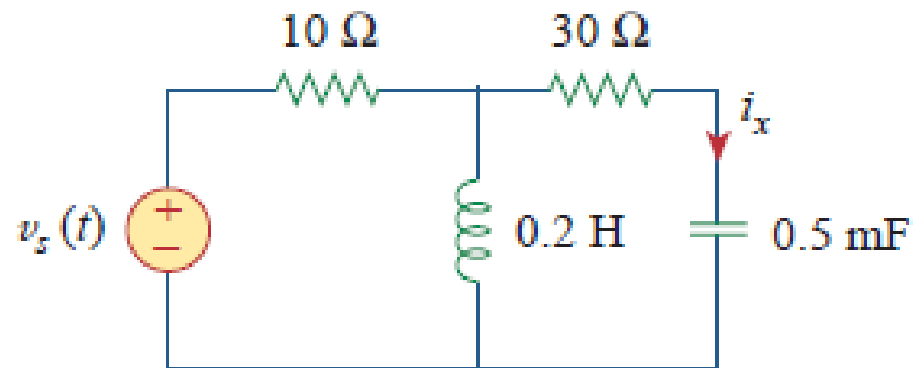
**Amaç:** Empedans matrisi ( $\mathbf{Z} \cdot \mathbf{I} = \mathbf{V}$ ) oluşturmak.

**Matris Kurulumu (Gözle Çözüm Tekniği):**



$$\begin{bmatrix} (4 + j2) & -(j2) \\ -(j2) & (2 + j2 - j5) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 20\angle 90^\circ \\ 0 \end{bmatrix}$$

Given that  $v_s(t) = 20 \sin(100t - 40^\circ)$  in Fig. 1, determine  $i_x(t)$ .



Applying KVL to mesh 1, we obtain

$$(8 + j10 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - j10\mathbf{I}_3 = 0$$

For mesh 2,

$$(4 - j2 - j2)\mathbf{I}_2 - (-j2)\mathbf{I}_1 - (-j2)\mathbf{I}_3 + 20\angle 90^\circ = 0 \quad (10.3.2)$$

For mesh 3,  $\mathbf{I}_3 = 5$ . Substituting this in Eqs. (10.3.1) and (10.3.2), we get

$$(8 + j8)\mathbf{I}_1 + j2\mathbf{I}_2 = j50 \quad (10.3.3)$$

$$j2\mathbf{I}_1 + (4 - j4)\mathbf{I}_2 = -j20 - j10 \quad (10.3.4)$$

Equations (10.3.3) and (10.3.4) can be put in matrix form as

$$\begin{bmatrix} 8 + j8 & j2 \\ j2 & 4 - j4 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} j50 \\ -j30 \end{bmatrix}$$

from which we obtain the determinants

$$\Delta = \begin{vmatrix} 8 + j8 & j2 \\ j2 & 4 - j4 \end{vmatrix} = 32(1 + j)(1 - j) + 4 = 68$$

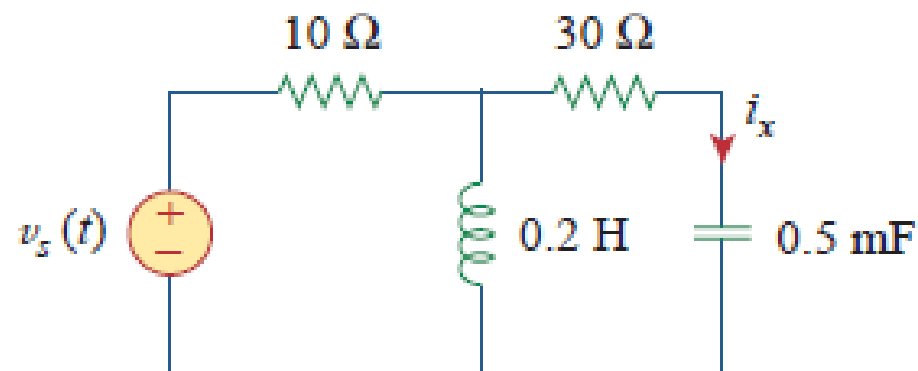
$$\Delta_2 = \begin{vmatrix} 8 + j8 & j50 \\ j2 & -j30 \end{vmatrix} = 340 - j240 = 416.17\angle -35.22^\circ$$

$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta} = \frac{416.17\angle -35.22^\circ}{68} = 6.12\angle -35.22^\circ \text{ A}$$

The desired current is

$$\mathbf{I}_o = -\mathbf{I}_2 = 6.12\angle 144.78^\circ \text{ A}$$

Given that  $v_s(t) = 20 \sin(100t - 40^\circ)$  in Fig. determine  $i_x(t)$ .



# Süperpozisyon Teoremi



**Teorem:** “Devrede birden fazla kaynak varsa, her birini tek tek devreye alıp sonuçları toplarız.”

## Senaryo 1: Aynı Frekanslı Kaynaklar



- **Kaynaklar:**  $10 \cos(10t)$  ve  $5 \sin(10t)$ .



**İşlem:** Fazör domaininde toplayabilirsiniz  
( $V_{\text{total}} = V_1 + V_2$ ).

## Senaryo 2: FARKLI Frekanslı Kaynaklar (DC + AC veya $\omega_1 \neq \omega_2$ )

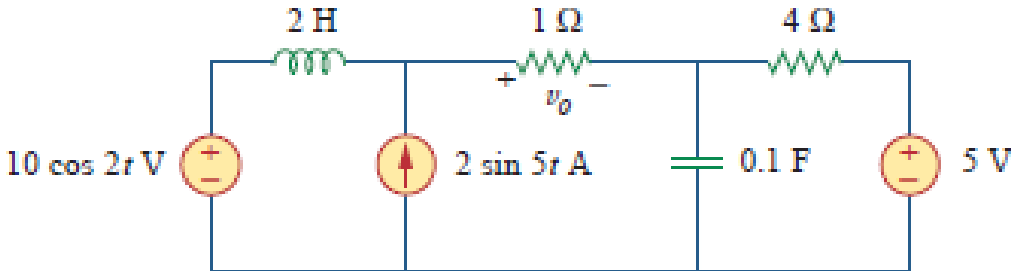


- **Kaynaklar:**  $10V$  (DC) ve  $5 \cos(50t)$ .



**Kritik Uyarı:** Doğrudan Fazör Toplamı YAPILAMAZ.  
Ayrı ayrı zaman domaininde çözülmelidir.

Find  $v_o$  of the circuit of Fig. 10.13 using the superposition theorem.



# Kaynak Dönüşümü (Source Transformation)



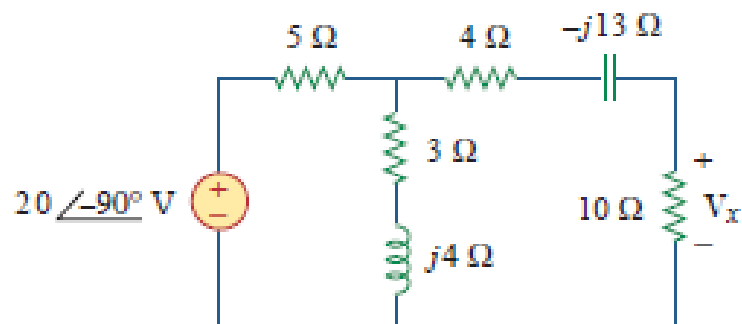
Kural: Bir Gerilim Kaynağı ( $V_s$ ) ve ona **SERİ** bağlı Empedans ( $Z_s$ ), bir Akım Kaynağı ( $I_s$ ) ve ona **PARALEL** bağlı Empedansa dönüşebilir.

Formül:

$$I_s = V_s / Z_s$$

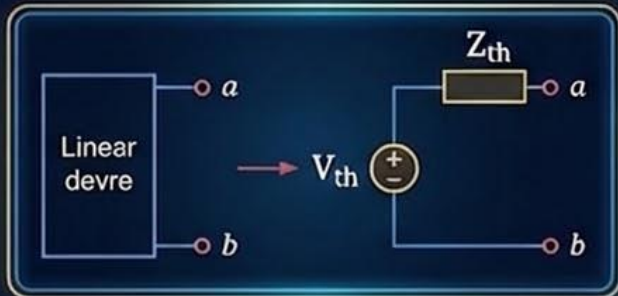
(Empedans değeri değişmez, sadece konumu değişir).

Calculate  $V_x$  in the circuit of Fig. 10.17 using the method of source transformation.



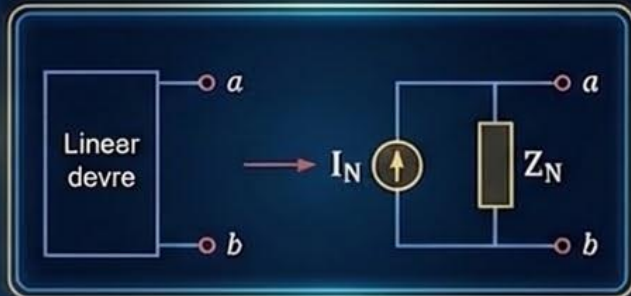
# Eşdeğer Devreler

## Thevenin Eşdeğer Devresi



- Karmaşık sayılar kullanılarak AC devrelere uygulanır.
- Lineer devre, Empedans ( $Z_{th}$ ) ile SERİ bağlı bir Gerilim Kaynağı ( $V_{th}$ ) ile değiştirilir.
- $V_{th}$  açık devre gerilimidir.

## Norton Eşdeğer Devresi



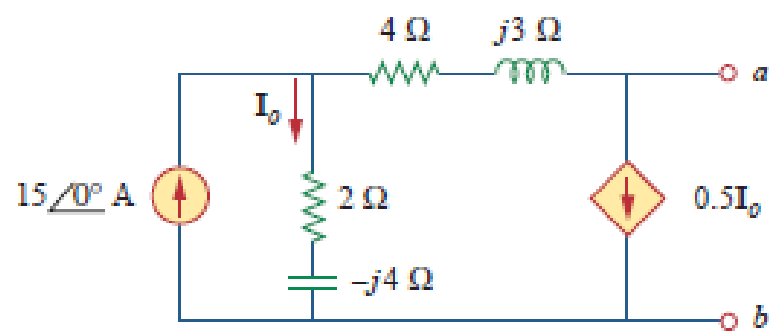
- Lineer devre, Empedans ( $Z_N$ ) ile PARALEL bağlı bir Akım Kaynağı ( $I_N$ ) ile değiştirilir.
- $I_N$  kısa devre akımıdır.

$$V_{Th} = Z_N * I_N$$
$$Z_{Th} = Z_N$$

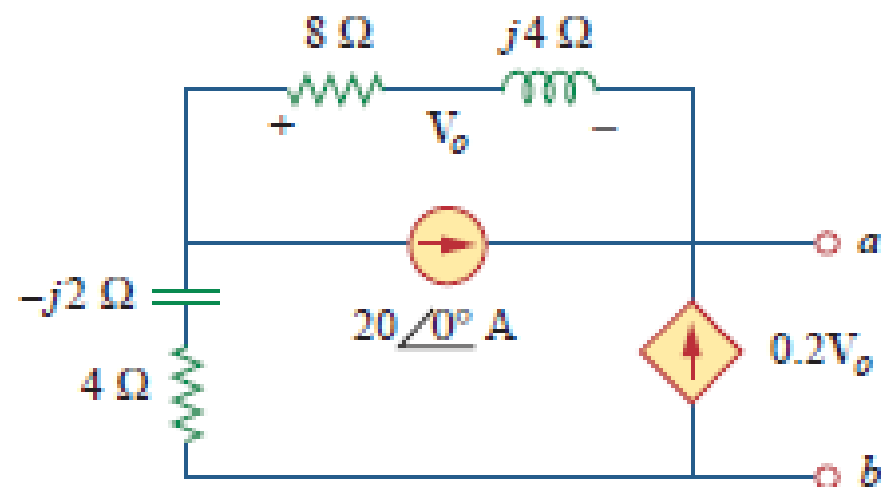


**Not:** Farklı frekanslarda çalışan kaynaklar için, eşdeğer devreler HER frekansta ayrı ayrı belirlenmelidir.

Find the Thevenin equivalent of the circuit in Fig. 10.25 as seen from terminals  $a$ - $b$ .



Determine the Thevenin equivalent of the circuit in Fig. 10.27 as seen from the terminals  $a$ - $b$ .



**Figure 10.27**

For Practice Prob. 10.9.